

Bayesian Nonparametrics and Bayesian Machine Learning – Students Seminar 2026

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1 Nature of the project

Students should form groups of two, each group undertaking one project. We suggest in Section **Papers List** a few scientific papers that can each lead to a project, but you can propose another paper. Acceptance of your proposal is subject to the instructor's approval.

For the selected paper, your group is required to undertake the following tasks:

1. Elucidate the Methodologies: Provide a comprehensive explanation of the theoretical, computational, and/or empirical methods employed in the paper.
2. Highlight Key Findings: Accentuate the principal conclusions and findings of the paper.
3. Practical Application or Extension: Apply the paper's methodologies or findings to a real-world dataset of your choosing, if applicable, or you may choose to apply them to an interesting simulated dataset. Another option is to extend the results, or to apply them to an unexpected or interesting special case.

Additional credit will be awarded for demonstrating creativity and critical thinking beyond the scope of the original paper. This could be in the form of a novel theoretical insight, a demonstrative experiment, or any other insightful addition that was not covered in the original paper. The objective is to engage with the paper critically and constructively.

2 Assignment of papers

As a first step, we ask each group to fill the spreadsheet at

<https://lite.framacalc.org/vqz151t4ld-9zl4>

with the title of the paper, a link to it (if available), and the composition of the group. We ask that you fill in the form **before 2026-02-24**.

3 Format of the deliverable

Please send:

- one report as a pdf (≤ 10 pages) in the [NeurIPS template](#),
- your presentation of at most 12 slides

to [your teacher](#)¹ **no later than March 6th 2026 at 23.59**. There will be no deadline extension, and **any delay will incur a 5 points penalty**.

The last step of your project will be a presentation in front of the class during the last lecture. The amount of time given to each group will depend on the number of students registered for the exam, but indicatively you should plan on 18 minutes (9 minutes each). Participation (asking questions during the other students' presentation) will also be evaluated.

¹ if the above link is broken, this means: guillaume.konkamking@inrae.fr

4 Evaluation criteria

Report and Slides quality

- 5: Structure is excellent
- 4: Structure is good, there are minor suggestions
- 3: Correct global organisation, but some details lack of clarity.
- 2: Navigating requires some effort. Lack of structure; some parts are confusing.
- 1: Student does not seem to have spent more than 5 mins on that
- 0: No slides

Clarity of the explanation/speech

- 5: Everything is as clear as blue sky, excellent
- 4: Clear overall, minor problems
- 3: Lack of clarity in details, but the idea explanation is understandable
- 2: Following requires some effort, some parts are confusing
- 1: Student doesn't seem to know what he/she is talking about
- 0: Don't know what to say. At all.

Understanding of the paper, answers to questions

- 5: Outstanding depth of analysis and validation process, excellent answers, insightful comments.
- 4: Very good depth of analysis, but some points remain to be investigated with more care.
- 3: Satisfying analysis, some potential mistakes or methodological errors in application, validation or interpretation.
- 2: Weaknesses in analysis, with some lack of depth.
- 1: Poor understanding of the topic.
- 0: Total misunderstanding.

Papers List

- [1] Michalis Titsias and Neil D. Lawrence. Bayesian gaussian process latent variable model. In Yee Whye Teh and Mike Titterton, editors, *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, volume 9 of *Proceedings of Machine Learning Research*, pages 844–851, Chia Laguna Resort, Sardinia, Italy, 13–15 May 2010. PMLR.
- [2] Christopher Yau, Omiros Papaspiliopoulos, Gareth O Roberts, and Christopher Holmes. Bayesian non-parametric hidden markov models with applications in genomics. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 73(1):37–57, 2011.
- [3] Nilesh Tripuraneni, Shixiang Shane Gu, Hong Ge, and Zoubin Ghahramani. Particle gibbs for infinite hidden markov models. *Advances in Neural Information Processing Systems*, 28, 2015.
- [4] Daniele Durante, David B Dunson, and Joshua T Vogelstein. Nonparametric bayes modeling of populations of networks. *Journal of the American Statistical Association*, 112(520):1516–1530, 2017.
- [5] Tamara Broderick, Ashia C. Wilson, and Michael I. Jordan. Posteriors, conjugacy, and exponential families for completely random measures. *Bernoulli*, 24(4B):3181 – 3221, 2018.
- [6] Fadhel Ayed, Juho Lee, and Francois Caron. Beyond the chinese restaurant and Pitman-Yor processes: Statistical models with double power-law behavior. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 395–404. PMLR, 09–15 Jun 2019.
- [7] Andrew Gordon Wilson. The case for bayesian deep learning. *arXiv preprint arXiv:2001.10995*, 2020.

- [8] Edwin Fong and Chris C. Holmes. Conformal bayesian computation. *Advances in Neural Information Processing Systems*, 34:18268–18279, 2021.
- [9] Liu Yang, Xuhui Meng, and George Em Karniadakis. B-pinns: Bayesian physics-informed neural networks for forward and inverse pde problems with noisy data. *Journal of Computational Physics*, 425:109913, 2021.
- [10] Filippo Ascolani, Antonio Lijoi, Giovanni Rebaudo, and Giacomo Zanella. Clustering consistency with Dirichlet process mixtures. *Biometrika*, 09 2022. asac051.
- [11] Laura D’Angelo, Antonio Canale, Zhaoxia Yu, and Michele Guindani. Bayesian nonparametric analysis for the detection of spikes in noisy calcium imaging data. *Biometrics*, 2022.
- [12] Ryan Giordano, Runjing Liu, Michael I Jordan, and Tamara Broderick. Evaluating sensitivity to the stick-breaking prior in bayesian nonparametrics. *Bayesian Analysis*, 1(1), 2022.
- [13] Sirio Legramanti, Tommaso Rigon, Daniele Durante, and David B Dunson. Extended stochastic block models with application to criminal networks. *The annals of applied statistics*, 16(4):2369, 2022.
- [14] Keith Levin and Brenda Betancourt. Fast generation of exchangeable sequence of clusters data. *arXiv preprint arXiv:2209.02844*, 2022.
- [15] Sébastien Marmin and Maurizio Filippone. Deep gaussian processes for calibration of computer models (with discussion). *Bayesian Analysis*, 17(4):1301–1350, 2022.
- [16] Lorenzo Masoero, Federico Camerlenghi, Stefano Favaro, and Tamara Broderick. More for less: predicting and maximizing genomic variant discovery via bayesian nonparametrics. *Biometrika*, 109(1):17–32, 2022.
- [17] Federica Zoe Ricci, Michele Guindani, and Erik B. Sudderth. Thinned random measures for sparse graphs with overlapping communities. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022.
- [18] Lorenzo Schiavon, Antonio Canale, and David Brian Dunson. Generalized infinite factorization models. *Biometrika*, 109(3):817–835, 01 2022.
- [19] Ba-Hien Tran, Simone Rossi, Dimitrios Milios, and Maurizio Filippone. All you need is a good functional prior for bayesian deep learning. *Journal of Machine Learning Research*, 23(74):1–56, 2022.
- [20] Arhit Chakrabarti, Yang Ni, Ellen Ruth A. Morris, Michael L. Salinas, Robert S. Chapkin, and Bani K. Mallick. Graphical dirichlet process, 2023.
- [21] Pierpaolo De Blasi and María F Gil-Leyva. Gibbs sampling for mixtures in order of appearance: the ordered allocation sampler. *Journal of Computational and Graphical Statistics*, 32(4):1416–1424, 2023.
- [22] Ethan Goan, Dimitri Perrin, Kerrie Mengersen, and Clinton Fookes. Piecewise deterministic markov processes for bayesian neural networks, 2023.
- [23] Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. Tabpfn: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations 2023*, 2023.
- [24] Servet Martínez and Javier Santibáñez. One step entropy variation in sequential sampling of species for the poisson-dirichlet process. *Acta Applicandae Mathematicae*, 184(1):6, 2023.
- [25] Michael Minyi Zhang, Bianca Dumitrascu, Sinead A Williamson, and Barbara E Engelhardt. Sequential gaussian processes for online learning of nonstationary functions. *IEEE Transactions on Signal Processing*, 71:1539–1550, 2023.

- [26] Gianni Franchi, Olivier Laurent, Maxence Leguery, Andrei Bursuc, Andrea Pilzer, and Angela Yao. Make me a bnn: A simple strategy for estimating bayesian uncertainty from pre-trained models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12194–12204, June 2024.
- [27] Vik Shirvaikar, Stephen G Walker, and Chris Holmes. A general framework for probabilistic model uncertainty. *arXiv preprint arXiv:2410.17108*, 2024.
- [28] Ziyu Wang and Chris Holmes. On subjective uncertainty quantification and calibration in natural language generation. *arXiv preprint arXiv:2406.05213*, 2024.
- [29] Dafne Zorzetto, Antonio Canale, Fabrizia Mealli, Francesca Dominici, and Falco J Bargagli-Stoffi. Bayesian nonparametrics for principal stratification with continuous post-treatment variables. *arXiv preprint arXiv:2405.17669*, 2024.
- [30] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, January 2025.
- [31] Alisa Sheinkman and Sara Wade. Understanding the trade-offs in accuracy and uncertainty quantification: Architecture and inference choices in bayesian neural networks. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 129–146. Springer, 2025.